

Quiz 4 - Computational Physics I

NAME: Juan Daniel Volsquez

SCORE: 16.8 / 20

Date: Thursday 29 May 2025 Duration: 45 minutes

Credits: 20 points (5 questions) Type of evaluation: LAB

Provide concise answers to the following items:

1. (4 points) Monte Carlo methods

(a) What do Monte Carlo methods consist of?

(b) What are the basic steps that you would follow to carry out a Monte Carlo simulation?

a) Monte Carlo methods are a special tool used to solve problems which have a large set of components and can be difficult to solve analytically, so we use M.C. to solve them in a statistical approach to get a really close approximation to solutions.

b) 1) Get the domain of the problem.
2) Set statistical distribution to generate random inputs.
3) Get the random numbers and calculate the needed values.
4) Aggregate results. deterministic step.
5) Check convergence in comparison with the expectation value.

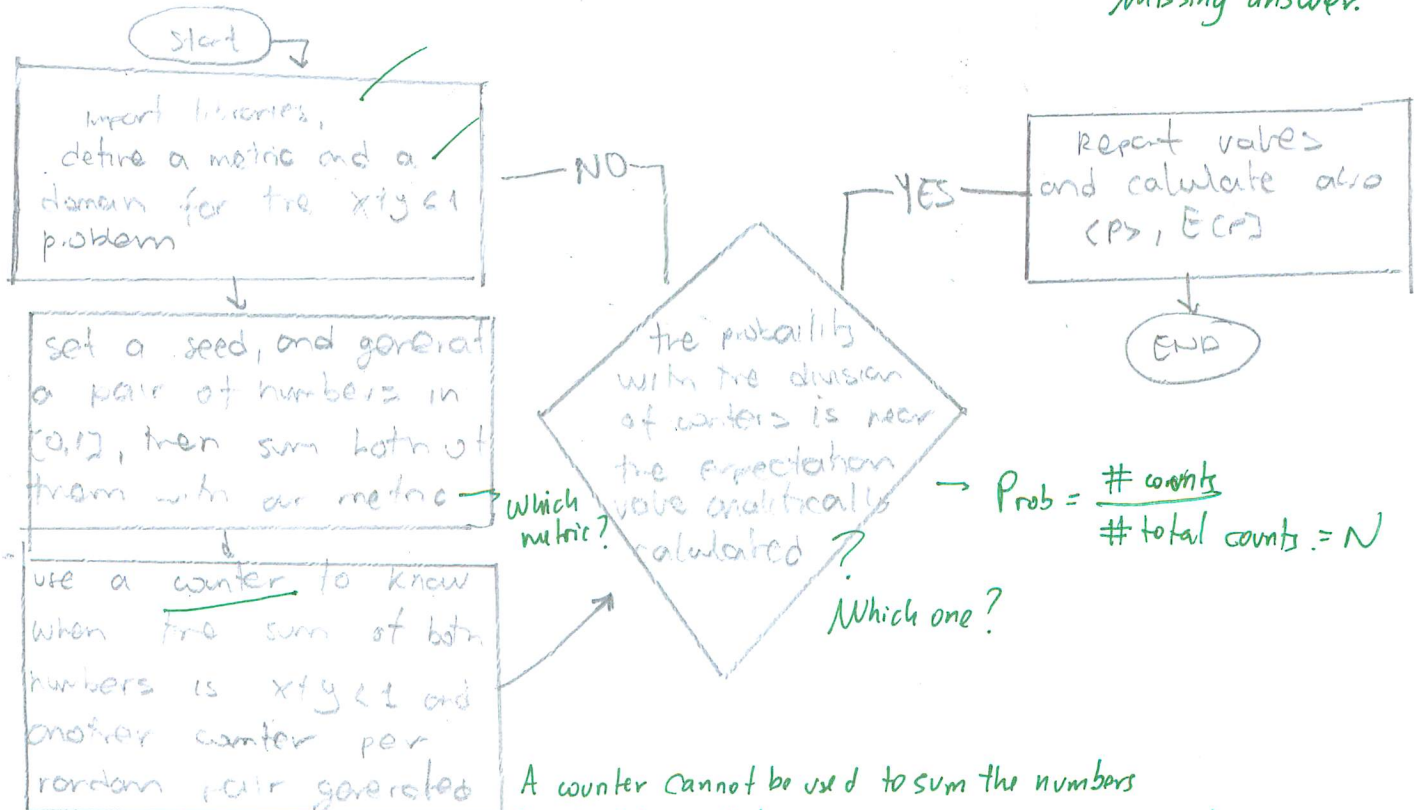
-0.25

2. (4 points) Monte Carlo simulation

Imagine you are asked to estimate the probability that the sum of two random numbers (x and y) drawn uniformly from the interval $[0, 1]$ is less than 1 (i.e., $x + y < 1$). Describe the python workflow that you would use to estimate the probability $P(x + y < 1)$. What value should it converge to?

-1.5

Missing answer.



A counter cannot be used to sum the numbers
It should count how many values are e.g. > 1

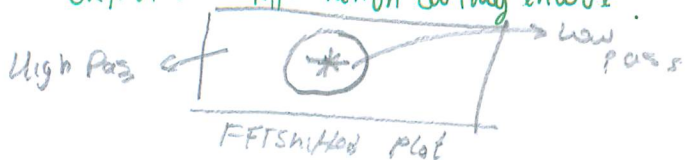
$$(x+y) > 1 \sim (x+y) \leq 1$$

3. (4 points) Fourier filters

- 0.5 (a) Explain the difference between a low-pass filter and a high-pass filter in Fourier analysis.
 (b) List the main steps you would take to detect edges in images using Fourier analysis in python.

a) For low pass filters we just take the low frequencies from a signal and for a High pass we just take high frequencies by filtering them on a usual distribution of the shifted FFT signal.

Ok, but what information do they encode?



- b) 1) FFT transform the image signal
 2) FFTshift the signal
 3) Plot the shifted signal
 4) Apply a High pass filter to keep edges and details and lose other info.
 5) De-shift and return to real space
 6) Plot and analyze edges

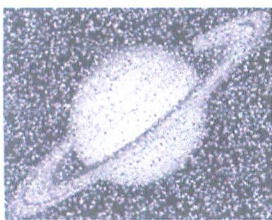
4. (4 points) Wavelet transforms

- (a) What is the difference between Fourier transforms and wavelet transforms?
 (b) Explain the process of wavelet decomposition for a 1D signal.

a) Fourier transform cannot manage time evolving signals or analyze a single signal on a wavelet or mixed signals data. wavelet transforms can.

b) we take a snapshot of the mixed signal and we apply a window function to separate and de-structure mixed signals on the wavelet and then obtaining the signal we want we can FT and analyze that one how? What is the result? $\rightarrow c_a$ c_d coef.

5. (4 points) Noise filtering



Imagine you are given the 2D grayscale image of Saturn with noise (on the left hand side). Describe the steps that you would take to remove the noise using a Fourier-based algorithm in python.

- 1) 2D FFT transform and shift the image
- 2) Since noise populates all frequencies, we plot on amplitudes the signal
- 3) Recognize the amplitudes correspondent to noise and generate a zero mask for them
- 4) Convolute signal and mask
- 5) Hopefully we have a denoised image